

Supplementary Material

BoxFusion: Reconstruction-Free Open-Vocabulary 3D Object Detection via Real-Time Multi-View Box Fusion

In this supplementary material, we present comprehensive technical details regarding our proposed approach, baseline implementations, and additional experimental results. Sec. 1 presents the implementation details, including both the procedure of our method and the optimization hyperparameters, to ensure full reproducibility of our work. Subsequently, we provide a detailed description of the experimental setup for baselines in Sec. 2, where we provide the details of implementation of data preprocessing to guarantee fair and comprehensive comparisons.

1. Implementation Details

In this section, we introduce the detailed setting of our method. As for the proposal generation using Cubify Anything [LGK*24], the threshold for the proposal generation is set to 0.4, and we also use a threshold to filter those proposals near the edges of the current image for invalid proposals. The generated proposals are then used for the subsequent steps of our method.

As for the 3D NMS thresholds for spatial association, we use $\tau = 0.1$ in both the CA-1M and ScanNetV2 datasets. For the multi-view box fusion, we use the pre-sampled particle swarm templates of 1024 particles, which are empirically found to be sufficient for the task. The particle swarm templates are sampled from a uniform distribution within the range of $[-1, 1]$ for the position dimension and $[-1, 1]$ for the dimension of box size. The random optimization is performed for up to 20 iterations, and it is terminated early if the objective function has already converged. More specifically, the initial search size of the center of PST is set to 0.1, and the initial search size of the box size is set to 0.5. The scaling coefficient for these two parameters is set to 0.1 and 0.5, respectively.

2. Details of Baselines

In this section, we describe the details of the preprocessing steps for point cloud and image-based input methods to ensure fair comparisons.

2.1. Point Cloud-Based Methods

For point cloud input methods, we compare against FCAF3D [RVK22], TR3D [RVK23], and SpatialLM [Tea25].

Since these methods require axis-aligned point cloud inputs, we perform the following adaptations.

- **CA-1M Dataset [LGK*24]**: The ground plane of the point cloud is already perpendicular to the z -axis. We further enforce axis alignment by ensuring walls are aligned with the principal axes through the following steps:

1. Detect dominant planes using RANSAC-based [FB81] plane fitting on the input scene point cloud.
2. Identify the largest plane orthogonal to the z -axis (assumed to be the dominant wall).
3. Rotate this plane to align with the x -axis, achieving full axis-alignment of the point cloud.

This preprocessing ensures compatibility with axis-aligned input requirements while preserving scene structure.

- **ScanNetV2 Dataset [DCS*17]**: axis-aligned point cloud is important for these point-based methods. We directly apply the official transformation matrices provided with the dataset to guarantee consistent axis alignment.

2.2. Image-Based Methods

For image-based methods, we evaluate two online instance segmentation approaches, since there are no similar online open-vocabulary object detection methods to be compared. The details are as follows:

- **EmbodiedSAM [XCZ*24]** uses FastSAM [ZDA*23] as its underlying image segmentation algorithm and aggregates the point clouds through its designed networks.
- **OnlineAnySeg [TZL*25]** applies CropFormer [QKS*23] for segmentation first and then fuses the multi-view masks in an online manner.

We employ their default configurations to obtain segmented point clouds. For fair comparison, we maintain each method's original frame sampling interval as specified in their respective implementations. Since these methods only provide segmented point clouds rather than oriented bounding boxes, we follow [DMNT21] for the post-processing steps to get corresponding oriented bounding boxes. The corresponding algorithm for obtaining bounding boxes is as shown in Tab. 1.

Table 1: Bounding Box Fitting for Image-Based Outputs

Oriented Bounding Box Extraction	
Input	Segmented point cloud P
Step 1. Preprocessing	Align point cloud via rigid transformation: Ground plane normal $\mathbf{n}_g$ aligns with $\mathbf{e}_z = [0, 0, 1]^T$
Step 2. Planar Projection	Project 3D points to xy -plane: $P_{2D} = \{(x_i, y_i) \mid \forall p_i \in P\}$
Step 3. Optimal Rotation Search	For $\theta_k \in [0, \pi)$ with step size $\Delta\theta$: • Rotate P_{2D} by θ_k: $\mathbf{R}_{\theta_k} = \begin{bmatrix} \cos \theta_k & -\sin \theta_k \\ \sin \theta_k & \cos \theta_k \end{bmatrix}$, $P_{2D}^{\theta_k} = \mathbf{R}_{\theta_k} P_{2D}$ • Compute axis-aligned bounding box (AABB): $x_{\min} = \min(x^{\theta_k}), \quad x_{\max} = \max(x^{\theta_k})$ $y_{\min} = \min(y^{\theta_k}), \quad y_{\max} = \max(y^{\theta_k})$ • Calculate area: $A(\theta_k) = (x_{\max} - x_{\min})(y_{\max} - y_{\min})$ Find $\theta^* = \arg \min_{\theta_k} A(\theta_k)$
Step4. Bounding Box Construction	• Final orientation: $\mathbf{o} = [\cos \theta^*, \sin \theta^*, 0]^T$ • z-dimension: $h = \max(z_i) - \min(z_i)$ • xy-dimensions: $w = (x_{\max} - x_{\min})_{\theta^*}, \quad d = (y_{\max} - y_{\min})_{\theta^*}$ • Center: $\mathbf{c} = \left[\frac{x_{\min} + x_{\max}}{2}, \frac{y_{\min} + y_{\max}}{2}, \frac{\min z_i + \max z_i}{2} \right]^T$ • Apply inverse transformation: $b_{\text{final}} = T_{\text{align}}^{-1} b_{\theta^*}$
Output	Oriented 3D bounding box $b_{\text{final}} = (\mathbf{c}, \mathbf{o}, w, d, h)$

References

- [DCS*17] DAI A., CHANG A. X., SAVVA M., HALBER M., FUNKHOUSER T., NIESSNER M.: Scannet: Richly-annotated 3d reconstructions of indoor scenes. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (2017), pp. 5828–5839. [1](#)
- [DMNT21] DHAMO H., MANHARDT F., NAVAB N., TOMBARI F.: Graph-to-3d: End-to-end generation and manipulation of 3d scenes using scene graphs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2021), pp. 16352–16361. [1](#)
- [FB81] FISCHLER M. A., BOLLES R. C.: Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM* 24, 6 (1981), 381–395. [1](#)
- [LGK*24] LAZAROW J., GRIFFITHS D., KOHAVI G., CRESPO F., DEHGHAN A.: Cubify anything: Scaling indoor 3d object detection. *arXiv preprint arXiv:2412.04458* (2024). [1](#)
- [QKS*23] QI L., KUEN J., SHEN T., GU J., LI W., GUO W., JIA J., LIN Z., YANG M.-H.: High quality entity segmentation. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)* (2023), IEEE, pp. 4024–4033. [1](#)
- [RVK22] RUKHOVICH D., VORONTSOVA A., KONUSHIN A.: Fcaf3d: Fully convolutional anchor-free 3d object detection. In *European Conference on Computer Vision* (2022), Springer, pp. 477–493. [1](#)
- [RVK23] RUKHOVICH D., VORONTSOVA A., KONUSHIN A.: Tr3d: Towards real-time indoor 3d object detection. In *2023 IEEE International Conference on Image Processing (ICIP)* (2023), IEEE, pp. 281–285. [1](#)
- [Tea25] TEAM M. R.: Spatiallm: Large language model for spatial understanding. <https://github.com/manycore-research/SpatialLM>, 2025. [1](#)
- [TZL*25] TANG Y., ZHANG J., LAN Y., GUO Y., DONG D., ZHU C., XU K.: Onlineanyseg: Online zero-shot 3d segmentation by visual foundation model guided 2d mask merging. *arXiv preprint arXiv:2503.01309* (2025). [1](#)
- [XCZ*24] XU X., CHEN H., ZHAO L., WANG Z., ZHOU J., LU J.: Embodiedsam: Online segment any 3d thing in real time. *arXiv preprint arXiv:2408.11811* (2024). [1](#)
- [ZDA*23] ZHAO X., DING W., AN Y., DU Y., YU T., LI M., TANG M., WANG J.: Fast segment anything. *arXiv preprint arXiv:2306.12156* (2023). [1](#)